

Jiatong Zhao

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Education

Shanghai Jiao Tong University, Zhiyuan Honors College

Sep. 2023 – Jun. 2027

Physics, Zhiyuan Honors Program

- GPA: 3.9/4.3; Average Core: 90.5/100; ranked 1/30 in major comprehensive evaluation.
- **Selected Coursework:** Machine Learning; Machine Learning Theory; CS224N: NLP with Deep Learning; CS336: Language Modeling from Scratch; Reinforcement Learning; ML Systems.
- **Honors and Awards:** 2025 MCM Meritorious Winner; Zhiyuan Honor Scholarship; Xiaomi Scholarship; Class of 2009 School of Electronic Information Alumni Scholarship; Category A Undergraduate Scholarship.

Research Experience

Training Stability for MoE RL: Router Stability and Train-Inference Mismatch

Oct. 2025 – Now

Advisor: Ning Ding, Tsinghua University; Ganqu Cui, Shanghai AI Lab

- Study how router dynamics and train-inference mismatch affect MoE training stability, expert specialization, and generalization.
- Analyze failure modes of router collapse and develop training recipes for more stable MoE learning.

Co4ICF: Physics-Informed Surrogate and RL-based Pulse Optimizer

Sep. 2025 – Feb. 2026

Supervisors: Junchi Yan and Jie Zhang

- Led the project and proposed Co4ICF (KDD 2026): physics-constrained surrogate + PPO for ICF laser pulse optimization with bi-frequency co-evolving calibration to mitigate OOD distribution shift.
- Achieved +46.1% yield gain under MULTI verification and about 990x optimization speedup; built a 90k-trajectory MULTI dataset for benchmarking.

Reinforcement Learning Framework for Diffusion Models with Generalized Reasoning Ability

Jul. 2025 – Oct. 2025

Advisor: Jie Fu, Shanghai AI Lab

- Proposed an RL framework for block-diffusion architectures.
- Explored whether a Turing-complete AR-diffusion architecture can acquire latent chain-of-thought reasoning purely through RL.

GeoBench: Rethinking Multimodal Geometric Problem-Solving via Hierarchical Evaluation

Mar. 2025 – May 2025

Advisor: Junchi Yan and Renqiu Xia, Shanghai Jiao Tong University

- Proposed a hierarchical benchmark for multimodal geometric problem-solving with four reasoning levels: visual perception, goal-oriented planning, rigorous theorem application, and self-reflective backtracking.
- Evaluated mathematical reasoning failures through structured formal-language signals; paper accepted by ICLR 2026.

Academic Papers

- [1] **Jiatong Zhao***, Tengyue Zhang*, Yuhan Wang*, Fuyuan Wu, Junchi Yan. “Co4ICF: Co-evolving Physics-Informed Surrogate and RL-based Pulse Optimizer for Inertial Confinement Fusion.” *SIGKDD 2026*, under review. **First-author work.**
- [2] Yuan Feng*, Yue Yang*, Xiaohan He*, **Jiatong Zhao**, Renqiu Xia, Bo Zhang, Junchi Yan. “GeoBench: Rethinking Multimodal Geometric Problem-Solving via Hierarchical Evaluation.” *ICLR 2026*, accepted.
- [3] Yufeng Zhao, Shenghe Zheng, Yun Luo, **Jiatong Zhao**, Qianjia Cheng, Fangchen Yu, Peng Ye, Ganqu Cui, Ning Ding, Yuan Shen, Kai Chen, Yu Cheng. “What Makes a Rubric Verifier Usable as an RL Reward? A Criterion-Vector Reward Interface for STEM Reasoning.” *EMNLP 2026*, under review.

Research Interests

I believe in *Physics of LM*: to underlying “Why” and to find the universal laws behind that. My preferred research paradigm is to start from concrete phenomena, formulate competing hypotheses, test them in controlled settings, and turn the resulting understanding into more reliable training and evaluation methods.

- **Reinforcement learning for reasoning:** improving the stability and robustness of RL-based reasoning systems under long-horizon generation, sparse feedback, and deployment-time constraints. Related readings: *Does RL Really Incentivize Reasoning?*; *When Speed Kills Stability*.
- **Understanding multimodal intelligence:** understanding how vision-language and generative models represent, simulate, and reason about real-world physical systems, why they fail, and how such insights can guide model, training, and data design.